MD MAHFUZUR RAHMAN

Norfolk, VA • +1 (757) 837-5456 • mrahm009@odu.edu • LinkedIn • Google Scholar • ORCiD • GitHub

Professional Summary

Engineering educator and researcher with 6+ years of university-level teaching experience across engineering fundamentals, instrumentation, electronics, digital signal processing, and laboratory-centered instruction. Experienced in core undergraduate engineering courses, project-based learning, curriculum development, and student assessment. Strong background in integrating computational tools (MATLAB, Python, LabVIEW) into engineering education. Committed to inclusive, student-centered teaching and academic success for diverse and first-generation student populations. Active researcher in biomedical instrumentation and noninvasive cardiovascular sensing, with 6 peer-reviewed journal publications (4 as first author) and 2 conference proceedings.

Education

Ph.D., Mechanical & Aerospace Engineering — Expected May 2026
Old Dominion University, Norfolk, VA | GPA: 3.87/4.0

- *Dissertation:* "Time-Frequency Analysis of Measured Arterial Pulse Signals for Motion Artifacts Removal"
- *Advisor:* Dr. Zhili Hao | NSF Grant #1936005

M.S., Applied Physics, Electronics & Communication Engineering — March 2012
University of Dhaka, Bangladesh | GPA: 3.73/4.0 (WES)

B.S., Applied Physics, Electronics & Communication Engineering — September 2010
University of Dhaka, Bangladesh | GPA: 3.81/4.0 (WES)

Teaching Interests

- Engineering Fundamentals and First-Year Engineering
- Engineering Instrumentation, Measurement, and Data Acquisition
- Biomedical Instrumentation and Sensors
- Signals and Systems / Digital Signal Processing
- Experimental Methods and Laboratory-Centered Instruction
- Programming-Supported Engineering Education (MATLAB, Python, LabVIEW)
- Project-Based Learning and Undergraduate Research Mentoring
- Student Retention and Success in Undergraduate Engineering

Academic Appointments

Graduate Teaching Assistant (Instructor) — Aug 2022 – Present
Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Graduate Teaching Assistant (Grader) — Aug 2019 – Present
Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Lecturer — Electronics & Telecommunication Engineering — Oct 2014 – Mar 2017
Atish Dipankar University of Science & Technology (ADUST), Dhaka, Bangladesh

Part-Time Lecturer — Aug 2016 – Mar 2017
Daffodil Institute of Information Technology (DIIT), Dhaka, Bangladesh

Teaching Experience

Graduate Teaching Assistant (Instructor) — ODU
August 2022 – Present

Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Courses Taught:

- MAE 203: Material Science Lab

Responsibilities:

- Designed and delivered laboratory-centered instruction aligned with course learning outcomes in materials science and engineering fundamentals
- Implemented project-based learning activities to reinforce core engineering concepts through hands-on experimentation and data analysis
- Developed laboratory reports and performance-based assessments to evaluate student learning outcomes
- Assessed student performance using formative and summative methods and refined instructional strategies based on assessment outcomes
- Coordinated laboratory content, instructional standards, and assessment practices across multiple course sections
- Utilized LMS platforms (Canvas/Blackboard) to manage course materials, assignments, grading, and student communication

Graduate Teaching Assistant (Grader) — ODU
September 2019 – Present

Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Courses Supported:

- MAE 434W: Project Design and Management I
- MAE 433: Mechanical Engineering Design II
- MAE 220: Solid Mechanics II
- MAE 205: Dynamics
- MAE 201: Materials Science
- MAE 111: Information Literacy and Research

Responsibilities:

- Supported large, multi-section undergraduate engineering courses ensuring consistent grading standards and assessment practices
- Assisted in development and evaluation of exams, quizzes, and project deliverables aligned with course learning objectives
- Conducted post-exam review sessions and mentored student teams to improve conceptual understanding and academic performance
- Provided instructional continuity during faculty absences through guest lectures and structured review sessions
- Administered end-of-semester course outcome evaluations on behalf of faculty in support of ABET accreditation documentation

Lecturer — Atish Dipankar University of Science & Technology
October 2014 – March 2017

Dept. of Electronics & Telecommunication Engineering (ETE), Dhaka, Bangladesh

Courses Taught:

- TPHY 108: Physics II with Lab
- ETE 208: Electrical Circuit I | ETE 209: Electrical Circuit II
- ETE 224: Electronics I | ETE 232: Electronics II | ETE 321: Electronics III
- ETE 222: Digital Logic Design
- ETE 229: Semiconductor Theory and Devices
- ETE 324: Microprocessor and Microcomputer
- ETE 331: Instrumentation and Measurement

- ETE 360: Digital Signal Processing
- ETE 401: Industrial and Power Electronics System
- ETE 415: Digital and Satellite Communication
- ETE 421: Wireless and Mobile Communication
- ETE 499: Project/Internship

Teaching & Instruction:

- Developed and delivered introductory and core undergraduate engineering courses in circuits, electronics, instrumentation, digital logic, microprocessors, and digital signal processing, with integrated laboratory components
- Designed outcome-driven syllabi, lecture materials, assignments, laboratory experiments, and examinations aligned with program learning objectives
- Integrated MATLAB-based signal analysis, numerical methods, and data visualization into electronics, instrumentation, and signal processing courses to support engineering problem-solving and experimental interpretation
- Applied project-based learning in laboratory and design-oriented courses to strengthen conceptual understanding and hands-on skills
- Implemented rubric-based grading and structured assessment methods to evaluate student learning outcomes and course effectiveness

Student Mentoring & Curriculum Development:

- Supervised undergraduate final-year projects and internships, guiding students through project planning, implementation, documentation, and technical presentation
- Provided academic mentoring and formative feedback to support students from diverse educational backgrounds
- Served on the department curriculum development committee, contributing to course sequencing, content alignment, and instructional consistency across multiple sections

Part-Time Lecturer — Daffodil Institute of Information Technology August 2016 – March 2017
Dept. of Computer Science and Engineering (CSE), Dhaka, Bangladesh

Courses Taught:

- CSE 123: Introduction to Electrical Engineering
- CSE 124: Introduction to Electrical Engineering Practical
- CSE 438: Microprocessor and Assembly Language

Responsibilities:

- Taught introductory engineering and computing courses with integrated laboratory components
- Delivered lectures and lab sessions in electrical engineering fundamentals and microprocessor systems
- Designed assessments and provided structured feedback to support student learning

Student Mentoring

- Md. Rajon Ahmed — Undergraduate Final-Year Internship Spring 2016
  Dept. of ETE, ADUST, Dhaka, Bangladesh
- Md. Kamrul Hasan — Undergraduate Final-Year Internship Spring 2016
  Dept. of ETE, ADUST, Dhaka, Bangladesh
- Md. Tauhidur Rahman — Undergraduate Final-Year Project Summer 2015
  Dept. of ETE, ADUST, Dhaka, Bangladesh
- Muhammad Ali — Undergraduate Final-Year Project Summer 2015
  Dept. of ETE, ADUST, Dhaka, Bangladesh
- Homayun Kabir — Undergraduate Final-Year Internship Summer 2015
  Dept. of ETE, ADUST, Dhaka, Bangladesh

Skills Summary

- **Programming & Computation:** MATLAB, Python, LabVIEW, SPSS; Machine Learning (graduate coursework - ECE 607)

- **Engineering & Design Tools:** AutoCAD, Autodesk Inventor, KiCad, Multisim
- **Data Acquisition:** LabVIEW-based DAQ (NI DAQ hardware), real-time signal monitoring and processing
- **Clinical Research Tools:** REDCap (data collection, entry, and quality management)
- **Laboratory & Instruction:** Experimental design, lab safety, hands-on instrumentation, rubric-based assessment
- **Learning Management:** Canvas, Blackboard; digital course delivery; learning-outcome-aligned assessment design

Research

Research Areas:

- Biomedical instrumentation and noninvasive cardiovascular measurement
- Time-frequency signal analysis and motion artifact characterization in physiological signals
- Soft polymer microfluidic tactile sensor design, fabrication, and clinical validation

Research Output: 6 peer-reviewed journal publications (4 as first author); 2 conference proceedings (both as first author); conducting NSF-funded research (Grant #1936005).

Publications

Journal Articles

- **Rahman, M.M.**, Toraskar, S., Hasan, M., & Hao, Z. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part I: Accelerometers and PPG Sensors. *Sensors*, 25(18), 5710. doi:10.3390/s25185710 [IF ≈ 3.5; SCIE; Q2]
- **Rahman, M.M.**, Toraskar, S., Hasan, M., & Hao, Z. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part II: Tactile Sensors. *Sensors*, 25(18), 5700. doi:10.3390/s25185700 [IF ≈ 3.5; SCIE; Q2]
- **Rahman, M.M.**, Hasan, M., & Hao, Z. (2025). Motion Artifacts (MA) At-Rest in Measured Arterial Pulse Signals: Time-Varying Amplitude in Each Harmonic and Non-Flat Harmonic-MA-Coupled Baseline. *Biosensors*, 15(9), 578. doi:10.3390/bios15090578 [IF ≈ 5.6; SCIE; Q1]
- **Rahman, M.M.**, Hasan, M., May, J.F., Herre, J.M., Reynolds, L., & Hao, Z. (in press). Radial and carotid arterial pulse signals for assessing cardiovascular function at rest and during post-exercise recovery in a heart transplant patient: a case study. *Medical Sensors & Imaging* (OP Publishing). doi:10.1088/2977-8425/ae4b40 [Accepted; New Journal]
- Toraskar, S., **Rahman, M.M.**, & Hao, Z. (2026). Pulse Measurement Alters the True Pulse Signal in an Artery: A Coupled String-SDOF Model. *ASME J. Engineering and Science in Medical Diagnostics and Therapy*, 9(1), 011207. doi:10.1115/1.4069099 [IF ≈ 1.1; ESCI; Q3]
- Hao, Z., **Rahman, M.M.**, Jutlah, L.B.L., Athaide, C.E., & Au, J.S. (2023). Axial Wall Displacement at the Common Carotid Artery is Associated With the Lamb Waves. *ASME J. Engineering and Science in Medical Diagnostics and Therapy*, 6(1), 011008. doi:10.1115/1.4056267 [IF ≈ 1.1; ESCI; Q3]
- **Rahman, M.M.**, Hasan, M., & Ullah, S.M. (2012). Design of a Low-Loss Y-Splitter for Optical Telecommunication Using a 2D Photonic Crystal. *International Journal of Computer Applications*, 60(14), 34–39. doi:10.5120/9763-3613

Conference Proceedings

- **Rahman, M.M.**, Twiddy, H., Reynolds, L., & Hao, Z. (2021). Measurement of Post-Exercise Response of Local Arterial Parameters Using an Adjustable Microfluidic Tactile Sensor. *IEEE EMBC 2021*, pp. 1284–1287. [Presented]. doi:10.1109/EMBC46164.2021.9630432
- **Rahman, M.M.**, Sultana, N.A., Vahala, L., Reynolds, L., & Hao, Z. (2020). Improved Vibration-Model-Based Analysis for Estimation of Arterial Parameters From Noninvasively Measured Arterial Pulse Signals. *ASME IMECE 2020*, V005T05A079. [Presented]. doi:10.1115/IMECE2020-24551

Honors & Awards

- Tiwari Endowed Graduate Scholarship, Old Dominion University 2021
- National Science & ICT (NSICT) Fellowship, Bangladesh 2010
- Merit-Based Scholarship, University of Dhaka 2008

Certifications

- Biomedical Research — Basic, CITI Program
  Issued March 2026 · Expires March 2029 · Credential ID: 72079387 *(Active)*
- Biomedical Research — Refresher 2, CITI Program
  Issued December 2023 · Expired December 2025 · Credential ID: 57666724
- Biomedical Research — Refresher 1, CITI Program
  Issued November 2021 · Expired November 2023 · Credential ID: 43904237
- Biomedical Responsible Conduct of Research, CITI Program
  Issued October 2020 · Credential ID: 39136829

Professional & Community Service

- **Conference Reviewer:** International Conference on Imaging, Vision & Pattern Recognition (IVPR)
- **Volunteer,** Virginia Medical Reserve Corps (MRC) May 2020–Present
  Volunteered in community public-health initiatives including COVID-19 vaccination clinics, community-based health screening programs, patient intake, and health event coordination

Industry Experience

Manager — Merchandising & Production Apr 2017 – Aug 2019
Afiya Knitwear Ltd., Savar, Bangladesh

- Directed end-to-end merchandising strategy and execution, from buyer negotiation and costing to production planning and shipment delivery
- Led cross-functional teams to ensure on-time delivery, cost control, and adherence to buyer quality and compliance standards
- Oversaw sourcing, sampling approvals, and risk mitigation for production delays or quality issues

References

Available upon request.
Primary academic reference: Dr. Zhili Hao, Professor, Dept. of Mechanical & Aerospace Engineering, Old Dominion University (zlhao@odu.edu)